输运 1. 动量 $\tau = -\eta \frac{du}{dy}$

2. 热量 $q = -\lambda \frac{dt}{dy}$

$[W/(m \cdot K)]$

α — 导温系数 m^2/s

3. 质量 $J = -D \frac{dc}{dy}$.

D — 扩散系数 m^2/s

流体：流体、连续介质模型、重度 $\gamma = \rho g$，质量体积 $v = 1/\rho$

$k_T = -\frac{1}{v}\left(\frac{\Delta v}{\Delta p}\right)_T$ $\alpha_v = \frac{1}{v}\left(\frac{\Delta v}{\Delta T}\right)_p$

☆粘性：

牛顿粘性定律：$\boxed{\tau_{yx} = \frac{F}{A} = -\eta \frac{du_x}{dy}}$

η $Pa \cdot s$ ν (m^2/s)

温度、压力影响粘度的系数

三、流体特性

1. 体积力、面积力、线张力

2. $\boxed{p = p_0 + \rho(w - w_0)}$

四、动

1. 欧 + 拉：四个比较

2. 稳定流动 vs 非稳

流线 迹线 流管 流束、有效截面流量 < 体 / 质

☆ 3. 流体流动连续性方程

$$\boxed{\frac{\partial \rho}{\partial t} = -\nabla\cdot(\rho\vec{u}) + r}$$

①稳定 ②二维/一维 ③压缩 ④有源

P19 $\begin{cases} B: \nabla(\rho\vec{u}) = r \qquad \nabla\cdot\vec{u} = \frac{r}{\rho} \\ C: 0 = \frac{-\partial(\rho\vec{u})}{\partial x} + 0 \\ D: \frac{\partial u}{\partial x} = 0 \end{cases}$

P20: A: $X - \frac{1}{\rho}\frac{\partial p}{\partial x} = \frac{\partial u_x}{\partial t} + u_x\cdot\frac{r}{\rho}$

B: $X - \frac{1}{\rho}\frac{\partial p}{\partial x} = u_x\frac{r}{\rho}$

C: $X - \frac{1}{\rho}\frac{\partial p}{\partial x} = 0$

$$\boxed{W - \frac{p}{\rho}\cdot\frac{u^2}{2} = \text{constant}}$$

$-W_R$ 阻

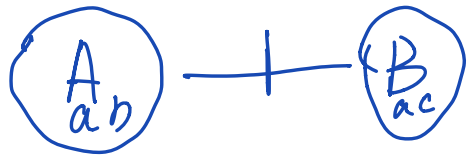
五.

1. Stefan-Maxwell.

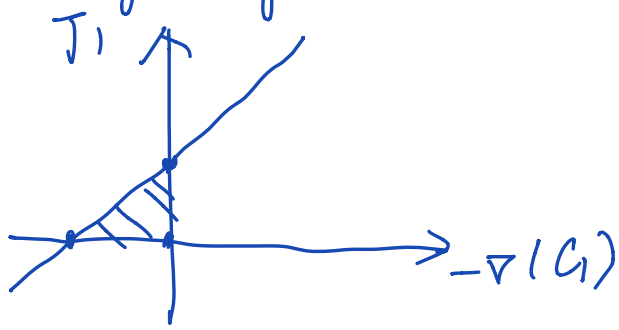
①克拉贝尔模型 $(-\nabla p(x_1)) = f_{12}\, x_1 x_2 (\vec{v}_1 - \vec{v}_2)$

$$\boxed{\nabla\left(\frac{p}{RT}x_i\right) = \frac{\vec{N}_2 x_1 - \vec{N}_1 x_2}{D_{12}}}$$

D_{12}:



γ_{ij} 与 D_{ij} 关系 ①②③④⑤



2、Knudsen 方程

① 实验模型

② $$\boxed{\nabla\left(\frac{P}{RT}x_i\right) = \frac{\vec{N}_i}{D_i^K}}$$

$$\boxed{Kn = \frac{\lambda}{d}}$$

3、Darcy 方程 $$\boxed{\vec{N}_i = -\frac{K}{\mu}\frac{P}{RT}x_i\nabla P}$$

K：渗透率 m^2

$Re = \frac{\rho v d}{\mu}$ 层流、紊流

六、热量传输

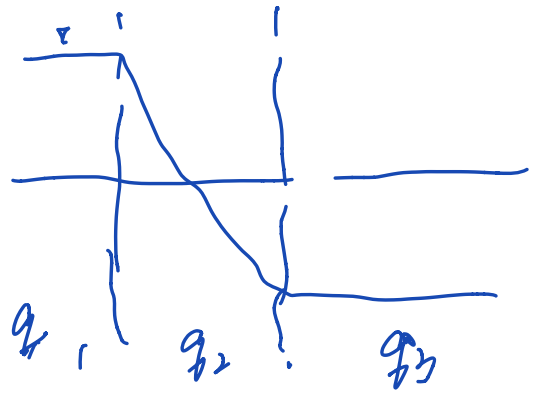
一、导热、对流、辐射.

二、温度场、温度梯度

$t = f(x, y, z, \tau)$.

等温面　　温度梯度

三、热流量 W　热流密度 $q = \frac{Q}{F}$ (W/m^2)　1、导热过程



$$q_1 = \frac{t_\infty - t_1}{R_{t_1}}$$

$$q_2 = \frac{t_1 - t_2}{R_{t_2}}$$

$$q_3 = \frac{t_2 - t_3}{R_{t_3}}$$

$$\boxed{q = q_1 + q_2 + q_3 = \frac{t_\infty - t_0}{R_{t_1} + R_{t_2} + R_{t_3}}}$$

1：稳态导热.

$$q = -\lambda \frac{\partial t}{\partial x}$$

λ $(W/m \cdot ^\circ C)$

$$\boxed{\lambda = \lambda_0 + \frac{T_1 + T_2}{2} b}$$

$$q = \frac{t_1 - t_{n+1}}{\underbrace{\sum_{i=1}^{n} \frac{\delta_i}{\lambda_i}}_{n\text{层平壁热阻}}} \quad W/m^2$$

热传导方程

$$\rho c \frac{\partial t}{\partial \tau} = \frac{\partial}{\partial x}\left(\lambda_x \frac{\partial t}{\partial x}\right) + \frac{\partial}{\partial y}\left(\lambda_y \frac{\partial t}{\partial y}\right) + \frac{\partial}{\partial z}\left(\lambda_z \frac{\partial t}{\partial z}\right) + q_v'$$

① 无相变

② $\lambda_x = \lambda_y = \lambda_z = \lambda$

③ $\rho c \frac{\partial t}{\partial \tau} = \frac{\partial}{\partial x} \lambda \left(\frac{\partial t^2}{\partial x^2} + \frac{\partial^2 t}{\partial y^2} + \frac{\partial^2 t}{\partial z^2}\right)$

④ $0 = \frac{\partial}{\partial x} \times \left(\lambda \frac{\partial t}{\partial x}\right)$

⑤ $0 = \frac{\partial^2 t}{\partial x^2} + \frac{\partial^2 t}{\partial y^2}$

三类边界条件

一. 二. 三

?

二维稳态导热解析解 含 exp. sin.

八. 非稳态

1. t升. 保持. 降

例题

2. $\theta = \frac{t - t_\infty}{t_0 - t_\infty} = \frac{\theta_t}{\theta_0}$

$Fo = \frac{a\tau}{L^2}$

$Bi = \frac{hL}{\lambda} = \frac{R_{\lambda}}{R_h}$ $L = \frac{V}{A}$ $Bi = \frac{L/\lambda}{1/h}$

$1/h \begin{cases} \frac{1}{h} \\ \frac{1}{h\Sigma} \end{cases}$

$W/(m^2 \cdot {}^\circ C)$

$Bi < 0.1$ MB $d\theta = h(t - t_\infty) A \, d\tau = -C_p M a t$

$$\boxed{\theta = \frac{t - t_\infty}{t_0 - t_\infty} = \exp(-Bi \, Fo)}$$

$Bi > 0.1$ MB

P55

$\rho c \frac{\partial t}{\partial \tau} = \frac{\partial}{\partial x}\left(\lambda \frac{\partial t}{\partial x}\right)$

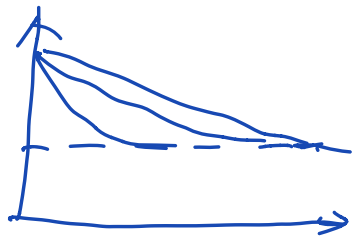
λ与t无关 $\frac{\partial t}{\partial \tau} = a \frac{\partial^2 t}{\partial x^2}$

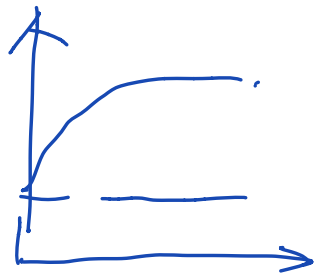
初: $t(x, 0) = t_0$

边: $\frac{\partial t}{\partial x}\Big|_{x=0} = 0$ $\frac{\partial t}{\partial x}\Big|_{x=\pm\delta} = h(t - t_\infty)$

结果 $\theta = f\left(Fo, Bi, \frac{x}{\delta}\right)$

$Bi \gg 0.1 M_B$





$$\frac{\partial t}{\partial \tau} = \alpha \frac{\partial^2 t}{\partial x^2}$$

初始：$t(x,0) = t_0$

边界：$\begin{cases} t(0,\tau) = t_\infty \\ \left.\frac{\partial t}{\partial x}\right|_{x=\infty} = 0 \end{cases}$

$$t = t(x,\tau) = t_0 + (t_\infty - t_0)\,\mathrm{erf}\left(\frac{x}{2\sqrt{a\tau}}\right)$$

退、正、淬

九、辐射换热

1、$\lambda < 0.38\mu m$

$0.38 < \lambda < 0.76$

$0.76 < \lambda < 1000$：

热射线

2、黑体 $Q = Q_A + Q_R + Q_D$

$1 = A + R + D$

灰体、绝对透热体

$\sigma T_0^4 \quad W/m^2$

3. $I_\lambda^\circ = \frac{dE^\circ}{d\lambda}$ W/m

图

4. $\lambda_m \times T = 2898$ (mm·K)

5. $\boxed{E^\circ = \sigma_0 T^4}$

6. 黑体辐射换热

$$\boxed{Q_{12}^\circ = \frac{\left(\frac{T_1}{100}\right)^4 - \left(\frac{T_2}{100}\right)^4}{\frac{1}{C_0 F}}}$$

7. 角系数

φ_{12} :

φ_{21} :

互换性原理 $\boxed{F_1\varphi_{12} = F_2\varphi_{21}}$

$$\boxed{Q_{12}^\circ = E_1^\circ F_1\varphi_{12} - E_2^\circ F_2\varphi_{21}}$$

8. 灰体

$$\boxed{\varepsilon = \frac{E}{E_0} \Rightarrow E = \varepsilon E_0 = \varepsilon C_0\left(\frac{T}{100}\right)^4}$$

有效辐射能力 $J = E + \rho G$

$= \varepsilon E^\circ + (1-A)G$

$\boxed{= \varepsilon E^\circ + (1-\varepsilon)G}$

$$G = \frac{J - \varepsilon E^{\circ}}{1-\varepsilon}$$

$$\boxed{Q_{12} = Q(-) = Q(+) = \frac{E_1^{\circ} - E_2^{\circ}}{\frac{1-\varepsilon_1}{\varepsilon_1 F_1} + \frac{1}{F_1 \varphi_{12}} + \frac{1-\varepsilon_2}{\varepsilon_2 F_2}}}$$

无限大 $F_1 = F_2 = F$ $\varphi_{12} = \varphi_{21} = 1$

$$\boxed{Q_{12} = \frac{E_1^{\circ} - E_2^{\circ}}{\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} - 1} \times F = C_{当}\left[\left(\frac{T_1}{100}\right)^4 - \left(\frac{T_2}{100}\right)^4\right] F}$$

$$C_{当} = \frac{C_0}{\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} - 1}$$

$$\left(\frac{1}{\varepsilon_1} - \frac{1}{\varepsilon}\right) + \left(\frac{1}{\varepsilon_2} - \frac{1}{\varepsilon}\right)$$

☆ + 隔热屏 $\left(\frac{1}{\varepsilon_1} - \frac{1}{\varepsilon}\right) + \left(\frac{1}{\varepsilon} - \frac{1}{\varepsilon}\right) + \left(\frac{1}{\varepsilon} - \frac{1}{\varepsilon}\right) + \left(\frac{1}{\varepsilon_2} - \frac{1}{\varepsilon}\right)$

$$\left(\frac{1}{\varepsilon} - \frac{1}{\varepsilon}\right) + \left(\frac{1}{\varepsilon} - \frac{1}{\varepsilon}\right)$$

$$C_{当}' = \frac{C_0}{\frac{1}{\varepsilon_1} + \varphi_{21}\left(\frac{1}{\varepsilon_2} - 1\right)}$$

十、综合题

1. 对+辐

$$Q = Q_{对} + Q_{辐} = \frac{t_1 - t_2}{\frac{1}{h\Sigma}} F$$

$$\boxed{q = \frac{t_g - t_a}{\frac{1}{h\Sigma_1} + \sum_{i=1}^{n} \frac{\delta_i}{\lambda_i} + \frac{1}{h\Sigma_2}}}$$

$$\boxed{h_c = \frac{-\lambda \frac{\partial t}{\partial y}\Big|_{y=0}}{t_f - t_s}}$$

十一、

1. 菲1 $\quad J = \frac{dM}{A\,dt} = -D\frac{\partial C}{\partial x}$

$$\frac{\partial C}{\partial \tau} = \frac{\partial}{\partial x}\left(D_x \frac{\partial C}{\partial x}\right) + \frac{\partial}{\partial y}\left(D_y \frac{\partial C}{\partial y}\right) + \frac{\partial}{\partial z}\left(D_z \frac{\partial C}{\partial z}\right)$$

1 Au-Ni 扩散偶

① Mo 固定

② Mo 不固定

$$\frac{\partial C_{Au}}{\partial \tau} = \frac{\partial}{\partial x}(-J_{Au})$$

$$\frac{\partial N_i}{\partial t} = \frac{\partial}{\partial x}(-J_{Ni})$$

$C_{Au} + C_{Ni}$ 常数.

$$\boxed{\tilde{D} = D_A r_B + D_B r_A}$$

十二、 C小 磷磁

基本假设.

$$J = -D\frac{\partial C}{\partial x}\Big|_{x=0} = \beta(C_p - C)$$

$\beta \to \infty$ 三→一

$$A + B\,\mathrm{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$\delta = 2k\sqrt{Dt} \qquad D = D_0 \exp\left(-\frac{Q}{RT}\right)$$

2. 2d P76.

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$

$$C(x,t) = \underbrace{\frac{M}{2\sqrt{\pi D t}}}_{\text{展宽}} \exp\left(-\frac{x^2}{4Dt}\right)$$

3: P78

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$

$\partial t \quad \partial x^2$

初: $C(x,0) = C_i$

边 $\begin{cases} \dfrac{\partial C(0,t)}{\partial x} = 0 & (x=0,\ t>0) \\ C(\pm L,\ t) = C_s & (x=\pm L,\ t>0) \end{cases}$

含exp 含cos

$$\theta = \frac{C - C_s}{C_i - C_s}$$